Git for collaborative work: illustration with exquisite corpse game

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This chapter puts into practice some of the core principles of Git seen previously.

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1 An exquisite corpse to discover collaborative work

So far, we have discovered the virtues of `Git` in an individual project. We will now go further with a collective project. As a reminder, in the previous chapter we covered the native concepts of `Git`: local and remote repositories (`remote`), the `clone` operation, *staging area*, `commit`, `push`, `pull`, and branches. We also covered some concepts related to the `Github` forge: authentication and *issues*.

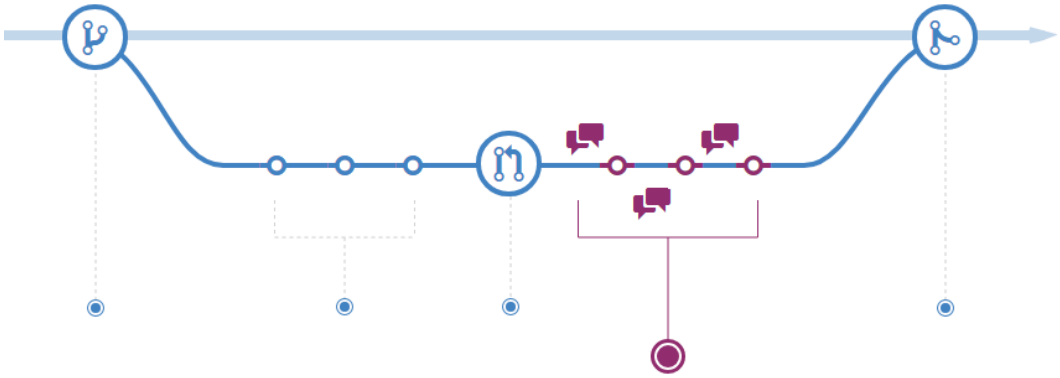
Now, we will discuss the challenges of collaborative work, which will lead us, in particular, to discuss the challenges of conflict management.

2 The *workflow* we adopt

We will adopt the simplest way of working, the *Github flow*. We already adopted it in the last exercise of the previous chapter, devoted to the use of branches.

The *Github flow* corresponds to this characteristic tree shape:

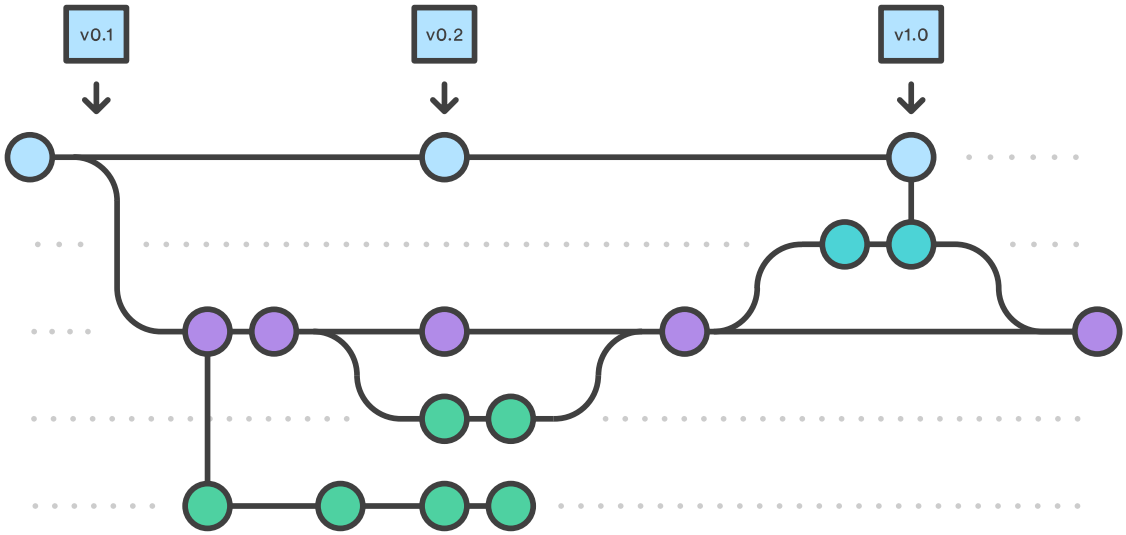
1. The `main` branch is the trunk
2. Branches start from `main` and diverge
3. When the changes are complete, they are merged into `main`; the branch in question disappears:



More complex *workflows* exist for large-scale projects.

There are more complex *workflows*, notably the `Git Flow` that I use to develop this course. This tutorial, very well done, illustrates with a graph the increased complexity of this flow:





This time, an intermediate branch, for example a `development` branch, gathers changes to be tested before integrating them into the official version (`main`).

> Tip
>
> You will find dozens of articles and books on this subject, each claiming to have found the best work organisation (*Git flow*, *GitHub flow*, *GitLab flow*...). Do not read too many of these books and articles or you will get lost (a bit like with magazines aimed at new parents...).
>
> The simplest way of working is the *Github flow* that we suggested you adopt. The tree is recognisable: branches diverge and systematically come back to `main`.
>
> For more complex projects in teams developing applications, other ways of working can be used, notably `Git flow`. There are no universal rules to determine the way of working; what matters, above all, is to agree on common working rules with your team.

2.1 Conflicts

During the last exercise of the previous chapter, we discovered, without really explaining it, the principle of a *merge*. Merging versions consists in reconciling two versions of a piece of code. This can be done by favouring one over the other or by choosing, in the passages that diverge, sometimes one and sometimes the other. We speak of conflicts to describe a situation where the same file has two different versions that must be reconciled.

`Git` greatly simplifies conflict management, which is one of the reasons for its success. If you have ever shared code by email in a team, you know that reconciling versions is extremely tedious: you have to check, more or less line by line, that the version you received by email does not differ from yours, which you have developed in parallel. Thanks to the fine-grained tracking of a file's evolution that `Git` allows, this conflict management is made easier. You will still have to favour one version over the other but it will be faster and more reliable.

While `Git` offers interesting features for conflict management, this is not an excuse to be disorganised. As we will see in the next exercises, we do have chapters

> Method for merges
>
> In the previous chapter, we *merged* our `issue-1` branch into `main`, our main branch. We went through the `Github` interface to do so. This is the recommended method for *merges* into `main`. It keeps an explicit trace of them (for example here), without having to search the sometimes complex tree of a project.
>
> Good practice is to make a `squash commit` to avoid an inflation of the number of commits in `main`: branches are meant to hold a multitude of small commits, while changes in `main` must be easy to trace, hence the idea of modifying small pieces of code.
>
> As we did in a previous exercise, it is very handy to add `close #xx` in the body of the message, where `xx` is the number of an *issue* associated with the `pull request`. When the `pull request` is merged, the *issue* will be closed automatically and a link will be created between the `issue` and the `pull request`. This will let you understand, several months or years later, how and why a given feature was implemented.
>
> On the other hand, integrating the latest changes from `main` into a branch is done locally. If your branch is in conflict, **the conflict must be resolved in the branch and not in main**. `main` must always remain clean.

2.2 History divergence: the different situations

2.2.a Simple case

Imagine two people collaborating on a project, Alice and Bob. Bob put the project aside for a few days. He wants to retrieve the progress Alice made during this period. She made the project evolve, let us assume with a single *commit* since it changes nothing.

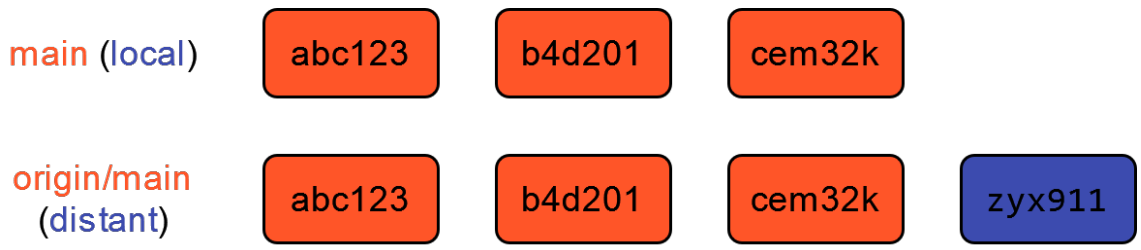


Figure 1: Bob's history when behind (simple case)

Bob's history is consistent with the one on `Github`, it is only missing the last *commit* in blue (Figure 1). Bob therefore just needs to retrieve this *commit* locally before starting to edit his files.

This type of merge is a *fast-forward merge*. The remote commit is added to the local history, without difficulty. The local repository is up to date with the remote repository again.

This is the ideal *merge* because it can be automated, and there is no risk of overwriting changes made by Bob with those of Alice. Adequate team organisation will make sure that most *merges* are of this type.

2.2.b A more complicated case

Now, imagine the more complicated case where Bob had made his code evolve in parallel, without retrieving Alice's changes before coming back to the project.

His local history therefore diverges from the remote history:

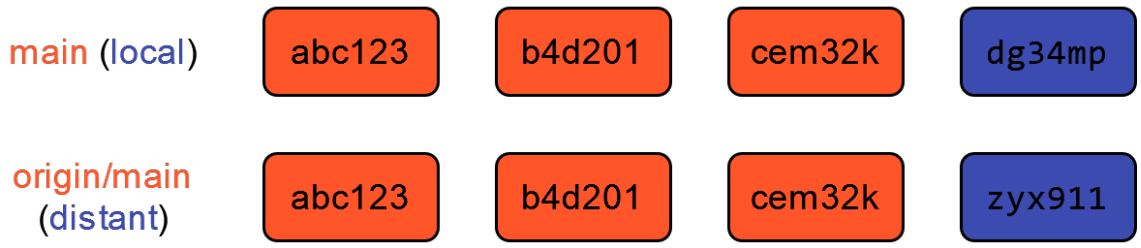


Figure 2: Bob's history when behind (more complicated case)

The last commits are not the same. `Git` cannot resolve the divergence by itself. It is up to Bob to decide which version he prefers. Two strategies are available to reconcile the histories:

- The *merge*;
- The *rebase*.

The first method, the simplest, is the *merge*.

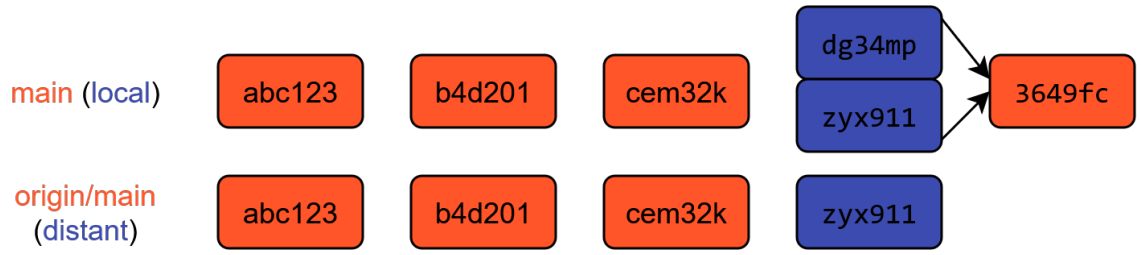


Figure 3: Bob's history when behind: resolution by *merge*

Bringing Alice's changes into Bob's history creates a first merge commit. The files in question will then contain markers identifying the version divergence and its source. In the raw file, this will give

```
import pandas as pd

<<<<<<< main
toto = pd.read_csv("source.csv")
=======
toto = pd.read_csv("source.csv", sep = ";")
>>>>>>> zyx911fhjehzafoldfkjknvjnvjnj;

toto.head(2)
```

`VSCode`, thanks to the `Git` extension, offers a native visualisation of this piece of code and offers, with a button click, different ways of reconciling the versions. It is of course always possible to edit these files: simply delete the markers and modify the lines in question.

```
import pandas as pd
Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
<<<<<<< main (Current Change)
toto = pd.read_csv("source.csv")
=======
toto = pd.read_csv("source.csv", sep = ";")
>>>>>>> zyx911fhjehzafoldfkjknvjnvjnj; (Incoming Change)
```

Figure 4: Conflicts in VSCode

Once the versions are reconciled, all that remains is to make a new *commit*. This history reconciles Alice's and Bob's versions. The drawback is that it makes the history non-linear (see the Atlassian documentation for more details) but it is `Git` that automatically handled this matter by creating a temporary branch. Until recently, this was the default behaviour of `Git`.

Another approach is available, the *rebase*. *In fine* this will correspond to this clean history.

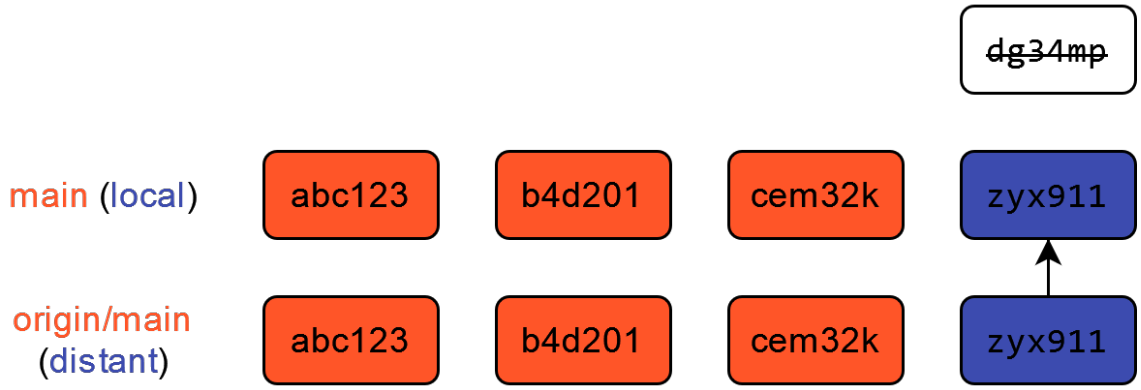


Figure 5

Nevertheless, this involves intermediate steps that consist in rewriting the history, which is already an advanced operation. In fact, `Git` performs three steps:

1. Temporarily removes the local *commit*
2. Performs a *fast forward merge* now that the local *commit* is no longer there
3. Adds the local commit back at the end of the history

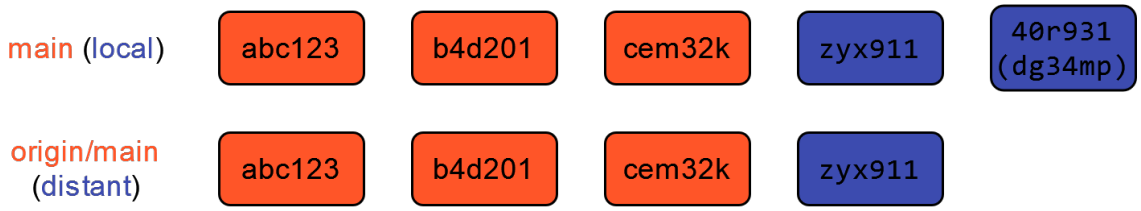


Figure 6

The local *commit* has changed identity, which explains its new SHA. This approach has the advantage of keeping a linear history. Nevertheless, it can have side effects and should therefore be used with caution (more explanations in the `Git` documentation). When you are starting out, and even

afterwards, it is recommended to favour the *merge* method, which is the one implemented by default. When you are more comfortable, you can try other *merge* methods.

3 Putting it into practice

Exercise 1: Interactions with the remote repository

This exercise is done in groups of three or four. There will be two roles in this scenario:

- One person will be responsible for being the **maintainer**
- Two to three people will be **developers**.

1️⃣ The maintainer creates a repository on `Github` **without clicking on the** `Add a README` **option**. Create a `.gitignore` using the `Python` template. They give rights to the project developer(s) (`Settings > Manage Access > Invite a collaborator`).

2️⃣ Each member of the project creates a local copy of the project (a *clone*). If you have a memory lapse, you can go back to the previous chapter to check the procedure.

3️⃣ Each member of the project creates a file with their last name and first name, following this structure `lastname-firstname.md` and avoiding special characters. They write three sentences of their choice in it **without punctuation or capital letters** (so that a correction can be made later). Finally, they *commit* on the project **through the visual extension**.

On the command line, the equivalent commands would be

```
git add lastname-firstname.md
git commit -m "This is the story of XXXXX"
```

4️⃣ Everyone tries to send (*push*) their local changes to the repository using the appropriate buttons of the visual extension.

On the command line, the equivalent command would be

```
git push origin main
```

5️⃣ At this stage, only one person (the fastest) should not have encountered a `push` rejection. This is normal: before accepting a change, `Git` first checks the consistency of the branch with the remote repository. The first person to have pushed modified the shared repository; the others must integrate these changes into their local version (*pull*) before being allowed to propose a change.

For those whose `push` was rejected, you will need to do a *pull*. Do it using the `VSCode` graphical extension.

On the command line, the equivalent command would be

```
git pull origin main
```

to try to bring the remote changes locally.

6️⃣ Look at the tree obtained in the interface to understand how the change from your teammate who could `push` earlier was integrated.

You will notice that your teammates' commits are integrated as they are into the history of the repository.

7️⃣ *Push* again. It is over for the 2nd person. The last person must redo steps 5 to 7 again (in a team of four it will have to be done once more).

❓ Question: what would have happened if the different members of the group had made their changes on one and the same file?

The next exercise offers an answer to this question.

Exercise 2: Handle conflicts when working on the same file

Continuing from the previous exercise, each person will work on the files of the other team members.

1️⃣ The two or three developers add the punctuation and capital letters to the first developer's file.

2️⃣ They skip a line and add a sentence (not all the same one).

3️⃣ Validate the results with a *commit* and make a *push*.

4️⃣ The fastest person normally encountered no difficulty (they can pause temporarily to watch what will happen to their neighbours). The others see their `push` rejected and must do a `pull`.

💥 There is a conflict, which should be signalled by a message like:

```
Auto-merging XXXXXX
CONFLICT (content): Merge conflict in XXXXXX.md
Automatic merge failed; fix conflicts and then commit the result.
```

5️⃣ Study the result of `git status`

6️⃣ If you open the offending files, you should see markers like

```
<<<<<<< HEAD
this is some content to mess with
content to append
=======
totally different content to merge later
>>>>>>> new_branch_to_merge_later
```

which are formatted by `VSCode` as in Figure 4.

7️⃣ Fix the files by choosing, for each block, the version that suits you thanks to the actions allowed by Git (`Accept Current Change`...)

Make a *commit* with the title "Conflict resolved by XXXX" where `XXXX` is your name.

8️⃣ Make a push. For the last person, redo operations 4 to 8

`Git` therefore makes it possible to work, at the same time, on the same file and to limit the number of manual actions needed to do the merge. When working on different parts of the same file, there is not even a need to make a manual modification, the merge can be automatic.

`Git` is a very powerful tool. But it does not replace good work organisation. As you have seen, this way of working solely on `main` can be painful. Branches make full sense in this case.

Exercise 3: Branch management

1️⃣ **The maintainer will contribute directly to `main` and does not create a branch. Each developer creates a local branch named `contrib-XXXXX` where `XXXXX` is their first name. To do so, in the `VSCode` interface, click on `... > Branch > Create branch` and enter the branch name in the appropriate menu**

On the command line, the equivalent command would be

```
git checkout -b contrib-XXXXX
```

2️⃣ Each member of the group creates a `README.md` file in which they write a subject-verb-object sentence. The maintainer is the only one to add a title to the `README` (which they *commit* to `main`).

3️⃣ Everyone *pushes* the product of their subconscious to the repository.

4️⃣ The developers each open a *pull request* on `Github`. The direction to choose is branch -> `main`. The developers give this *pull request* an explicit title.

The maintainer chooses one of the `pull request`s and validates it with the `squash commits` option. Check the result on the home page.

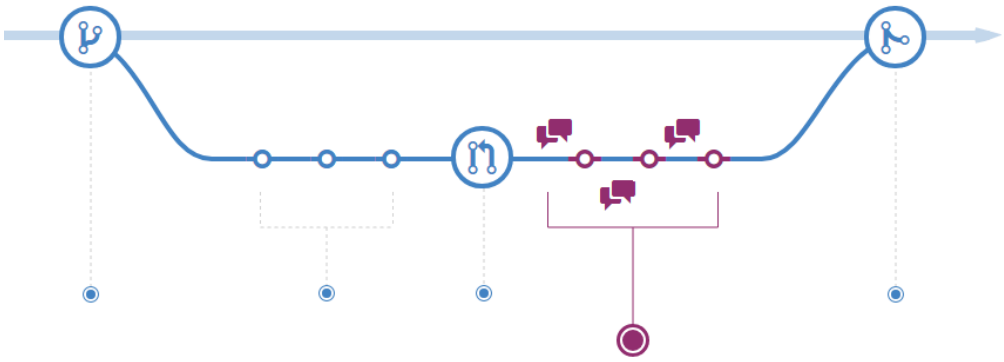
5️⃣ Each developer goes back to `main` by clicking on `... > Checkout to` and choosing `main`. Do a *pull*.

6️⃣ Go back to your branch and do `... > Branch > Merge` and choose `main`. Look at the resulting tree.

7️⃣ The author (if 2 developers) or the two authors (if 3 developers) of the non-validated `pull request` must repeat operations 5 and 6.

8️⃣ Once the version conflict is resolved and pushed, the maintainer validates the `pull request` following the same procedure as before.

9️⃣ Check the tree of the repository in `Insights > Network`. Your tree must have the characteristic shape of what is called the `Github flow`:



It is absolutely not mandatory for every collaborative project to choose this way of collaborating. For many projects, where people do not edit the same part of a file at the same time, going directly through `main` is enough. The mode above is important for substantial projects, where the `main` branch must be impeccable because, for example, it triggers a series of automated tests and the automated deployment of a deliverable. But for more modest projects, it is not necessary to go to extreme formalism. Good use of `Git` is pragmatic use, where it is used for its advantages and where work organisation adapts to them.

4 The specific challenges of the difficult interaction between `Git` and *notebooks*

The *notebook* format is very interesting for experimentation and for the final dissemination of results. Nevertheless, the complex structure of a *notebook* makes version control complicated. Indeed, opening it in a suitable editor (`Jupyter` or `VSCode`) provides a rendering that does not match the raw structure of the file as stored on disk. Behind the scenes, *notebooks* are JSON files that embed many elements: code, execution results, additional metadata... The fine-grained tracking of file changes that `Git` allows is complicated on a file with such a structure. The goal of this exercise is to illustrate these challenges and to mention some possible solutions.

Exercise 4: Version control and notebooks, an uneasy marriage

This exercise is still done as a team. When asked to make a *commit*, give it a meaningful name so that you can easily find your way when asked to look at the history. Do not *push* or *pull* until you get to the part where this is requested.

1. All team members go back to their `main` branch on their working copy

Each team member does the following questions.

2. Measure the size of the whole history on the command line with the command

```
du -sh .git
```

3. Download the *notebook* used for this exercise before the following command, on the command line:

```
curl "https://minio.lab.sspcloud.fr/lgaliana/python-ENSAE/inputs/git/exemple.ipynb" -o "notebook.ipynb"
```

- Do not open this file (the subject of the next question).
- Make a *commit* of this *notebook* then redo

```
du -sh .git
```

- Observe the first change in the size of our history

4. Open the *notebook*, run its cells, reordering them if necessary to debug the file.

- Make a *commit* of this *notebook* when it works then redo

```
du -sh .git
```

5. Now, make the following changes to the file:
   - Change the `df.head(3)` cell to `df.head(20)`
   - Create a new cell with the code `df["TYPEQU"].value_counts().plot(kind = "bar")`
   - Change the `zoom_start` variable in the cell producing the interactive map. Set its value to 13 instead of 15.
- Make a *commit* of this *notebook* when it works then redo

```
du -sh .git
```

6. Change the colour of the *barplot* with the `color` argument. Save, commit then redo

```
du -sh .git
```

7. Look at the history from the `VSCode` extension. Observe how your file evolves at each *commit*.

Now, we can move on to the collaborative step.

8. Each team member *pushes*:

- If the *push* works (the fastest person), modify the colour of the *barplot* again. *Commit* but do not *push* (wait until the other group members have done so then move on to question 9)
- If the *push* does not work (the slower people), move on to question 9.

9. In the `VSCode` extension, display the tree by clicking on the `View Git Graph` button.

- At the top of it, click on `Uncommitted changes`. This displays the changes that are not yet accepted.
- Click on `notebook.ipynb` to see the *diff* between your version and your teammate's. Do you understand the problem?
- Manually accepting each difference would be too costly. Accept your teammate's version (*Accept incoming*). *Commit* and observe the *diff*. Push.

For the last person in the group who could not push, redo question 9. After pushing, the person who was fastest in question 8 can do question 9 by retrieving the latest *commits*.

This exercise illustrates three points of difficulty with *notebooks*:

- The size of `ipynb` files quickly becomes large because the *outputs* are inserted into them, in raw form.
- It quickly becomes difficult to follow code changes because the code is drowned among other changes (notably those of the *outputs*).
- Conflict resolution is complicated because the JSON structure has to be preserved. It is in fact common for a *merge* to break this structure and make the *notebook* unreadable.

To solve these problems, there are several methods:

- If you want to keep the *notebook* as the place where code is stored, you must click on the `Clear all outputs` button before a `git add` step. The `nbstripout` *package* can be an interesting solution to avoid doing this manually each time, as you risk forgetting and thus making the repository size grow anyway. Nevertheless, this approach only solves part of the problems since it only lightens the versioned JSON; it makes the risk of breaking the *notebook* during conflict resolution less likely but does not make it impossible.
- Work in a team on different *notebooks*, as was advised for text files. This avoids conflicts altogether. However, it risks inducing code redundancy and therefore problems later on if you want to synthesise the different pieces of work.
- Adopt a more modular structure by moving as much code as possible into `.py` files and importing the appropriate elements into the notebooks. This allows you to take advantage of `Git`, which tracks `.py` files very well, while also providing benefits for the quality of the *notebooks*. Being less monolithic, they will probably be of better quality.

This last approach is a way of opening up to the good practices discussed in the 3rd-year course on "Putting into production" (in French).

5 Conclusion

These chapters devoted to `Git` have helped demystify this software by illustrating, through practice, the main concepts and the daily routine. They aim to save you precious hours because learning `Git` on your own is often frustrating and incomplete. It is essential to keep the habit of using `Git` on your projects. They will be of better quality.

We have only seen the basic features of `Git` and `Github`. The purpose of the ENSAE 3rd-year course "Putting *data science* projects into production" (in French) is to introduce other features of `Git` and `Github` that make it possible to produce more ambitious, more reliable and more scalable projects.

Informations additionnelles

i Python environment

This site was built automatically through a `Github` action using the `Quarto` reproducible publishing software (version 1.10.18).

The environment used to obtain the results is reproducible via uv. The `pyproject.toml` file used to build this environment is available on the `linogaliana/python-datascientist` repository

```
[project]
name = "python-datascientist"
version = "0.1.0"
description = "Source code for Lino Galiana's Python for data science course"
readme = "README.md"
requires-python = "≥3.13,<3.14"
dependencies = [
    "altair≥6.0.0",
    "cartiflette≥0.2",
    "contextily==1.6.2",
    "duckdb≥0.10.1",
    "folium≥0.19.6",
    "gdal==3.11.4",
    "graphviz==0.20.3",
    "great-tables≥0.12.0",
    "gt-extras≥0.0.8",
    "ipykernel≥6.29.5",
    "jupyter≥1.1.1",
    "jupyter-cache≥1.0.0",
    "kaleido≥0.2.1",
    "langchain-community≥0.3.27",
    "loguru==0.7.3",
    "markdown≥3.8",
    "nbclient≥0.10.0",
    "nbformat≥5.10.4",
    "nltk≥3.9.1",
    "openpyxl≥3.1.5",
    "pandas≥3.0",
    "pip≥25.1.1",
    "plotly≥6.1.2",
    "plotnine≥0.15",
    "polars≥1.8.2",
    "py7zr≥1.1.3",
    "pyarrow≥17.0.0",
```

```
    "pynsee≥0.1.8",
    "python-dotenv≥1.0.1",
    "python-frontmatter≥1.1.0",
    "pywaffle≥1.1.1",
    "requests≥2.32.3",
    "scikit-image≥0.24.0",
    "scikit-learn≥1.8.0",
    "scipy≥1.13.0",
    "seaborn≥0.13.2",
    "selenium<4.39.0",
    "spacy≥3.8.4",
    "webdriver-manager≥4.0.2",
    "wordcloud==1.9.3",
]

[tool.uv.sources]
gdal = [
  { index = "gdal-wheels", marker = "sys_platform == 'linux'" },
  { index = "geospatial_wheels", marker = "sys_platform == 'win32'" },
]

[[tool.uv.index]]
name = "geospatial_wheels"
url = "https://nathanjmcdougall.github.io/geospatial-wheels-index/"
explicit = true

[[tool.uv.index]]
name = "gdal-wheels"
url = "https://gitlab.com/api/v4/projects/61637378/packages/pypi/simple"
explicit = true

[dependency-groups]
dev = [
    "nb-clean≥4.0.1",
]
```

To use exactly the same environment (version of `Python` and *packages*), please refer to the documentation for `uv`.